

Enhanced audio-visual speech enhancement with posterior sampling methods in recurrent variational autoencoders

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ABSTRACT

Recovering intelligible speech in noise is essential for robust communication. This work presents an audio-visual speech enhancement framework based on a Recurrent Variational Autoencoder (AV-RVAE), where posterior inference is extended using sampling-based methods including the Metropolis-Adjusted Langevin Algorithm (MALA), Langevin Dynamics EM (LDEM), Hamiltonian Monte Carlo (HMC), Barker sampling, and a hybrid MALA+Barker variant. To isolate the contribution of visual cues, an audio-only baseline (A-RVAE) is trained and evaluated under identical data and inference conditions.

Performance is assessed using Scale-Invariant Signal-to-Distortion Ratio (SI-SDR), Perceptual Evaluation of Speech Quality (PESQ), and Short-Time Objective Intelligibility (STOI), along with anytime convergence curves (metric versus wall-clock time) and the Real-Time Factor (RTF; ratio of runtime to audio duration) to measure computational efficiency.

Experimental results show that the hybrid MALA+Barker sampler achieves the best overall performance, while LDEM and step-size-optimized MALA exhibit the lowest RTFs, the MALA+Barker sampler offers the most favorable balance between efficiency and enhancement quality. Across all sampling strategies, the AV-RVAE consistently surpasses the audio-only baseline, particularly at low SNRs, confirming the benefit of visual fusion combined with advanced posterior sampling for robust speech enhancement in challenging acoustic environments.

1. Introduction

Enhancing speech quality and intelligibility in noisy environments is essential for effective communication in telecommunications, hearing aids, automatic speech recognition (ASR), and human-computer interaction. Traditional speech enhancement techniques, such as spectral subtraction (Boll, 1979), Wiener filtering (Lim and Oppenheim, 1979), and hidden Markov models (HMMs) (Veisi and Sameti, 2013), primarily utilize audio data. However, these methods typically struggle in highly noisy, non-stationary conditions or environments with very low signal-to-noise ratios (SNRs) (Vincent et al., 2018), motivating the shift toward data-driven approaches.

Recent advancements in deep learning have significantly improved speech enhancement by enabling models to learn complex relationships between noisy and clean speech signals (Wang et al., 2014; Mohammadiha et al., 2013). Supervised learning methods, notably deep neural networks (DNNs) (Wang and Chen, 2018; Yen et al., 2023; Richter et al., 2023) and convolutional neural networks (CNNs) (Karthik and Iqbal, 2023), achieve excellent performance but depend heavily on large datasets of labeled noisy-clean speech pairs. This reliance on labeled data limits their adaptability to diverse and unobserved noise conditions encountered in real-world applications.

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In response, unsupervised learning techniques have gained attention, particularly Variational Autoencoders (VAEs) (Kingma and Welling, 2014; Bando et al., 2018; Leglaive et al., 2018). VAEs model the distribution of clean speech without relying on paired datasets, enhancing generalization to unseen noise environments. Extensions such as Dynamic Variational Autoencoders (DVAEs), including the recurrent variational autoencoder (RVAE), incorporate temporal modeling through recurrent architectures (Leglaive et al., 2020; Girin et al., 2021), effectively capturing both spectral and temporal structures in speech. While Girin et al. (2021) provides a comprehensive theoretical overview of dynamical VAEs, it is important to acknowledge that RVAE was first applied to speech enhancement in Leglaive et al. (2020), with subsequent extensions introduced in Bie et al. (2022). Despite these advantages, most VAE-based approaches remain limited to audio-only scenarios, restricting their effectiveness under extreme noise conditions.

Audio-visual speech enhancement (AVSE) addresses these limitations by integrating visual cues, such as lip movements, to supplement auditory data (Michelsanti et al., 2021; Gabbay et al., 2018). Visual information significantly aids in resolving acoustic ambiguities, enhancing speech intelligibility in challenging environments. Nonetheless, AVSE presents unique challenges, notably the *McGurk effect* (McGurk and MacDonald, 1976), a perceptual phenomenon where conflicting audio-visual cues produce perceptual distortions. Recent approaches address these challenges through adaptive modality weighting (Magrin-Chagnolleau et al., 2008; Papandreou et al., 2009) and enhanced robustness to perceptual conflicts (Getz and Toscano, 2021). The first AV-VAE model integrating audio and visual modalities using NMF-based noise modeling was proposed in Sadeghi et al. (2020), which strongly inspires the present work.

In our previous research, we introduced an Audio-Visual Recurrent Variational Autoencoder (AV-RVAE) (Foroushi and Dansereau, 2024), extending the RVAE model by incorporating visual data for unsupervised speech enhancement. The AV-RVAE significantly outperformed existing audio-only models and audio-visual dynamic Kalman filter-based methods (Golmakani et al., 2023) across standard performance metrics, such as Scale-Invariant Signal-to-Distortion Ratio (SI-SDR) (Roux et al., 2019), Perceptual Evaluation of Speech Quality (PESQ) (Rix et al., 2001), and Short-Time Objective Intelligibility (STOI) (Taal et al., 2011). However, the inference process in noisy conditions remains challenging, as traditional variational inference often fails to fully capture complex posterior distributions, reducing enhancement effectiveness.

Recent work has introduced posterior sampling methods — such as Metropolis–Hastings EM (MH-EM), Langevin Dynamics EM (LDEM), and the Metropolis-Adjusted Langevin Algorithm (MALA) (Sadeghi and Serizel, 2024) — to improve inference accuracy in audio-only speech enhancement. These approaches offer more expressive posterior approximations than traditional variational methods.

Building on this foundation, our study extends posterior sampling to the audio-visual domain by integrating Hamiltonian Monte Carlo (HMC) (Chen and Gatmiry, 2023; Neal, 2012), Barker sampling (Livingstone and Zanella, 2022), and a novel hybrid MALA-Barker approach within the AV-RVAE framework. To the best of our knowledge, these methods have not previously been explored for audio-visual speech enhancement. Their application provides both methodological advances and practical benefits in terms of inference robustness, enhancement quality, and computational efficiency under challenging noise conditions.

The main contributions of this paper are:

- We extend the AV-RVAE framework by integrating advanced posterior sampling algorithms — including MALA, HMC, LDEM, and Barker — to improve latent variable inference in audio-visual speech enhancement.
- We provide a detailed empirical comparison of these inference strategies in terms of enhancement performance (SI-SDR, PESQ, STOI) and computational efficiency (RTF), showing that sampling-based methods outperform standard MAP-EM inference.
- We demonstrate the benefits of combining audio and visual modalities in challenging noise conditions, where AV-RVAE with posterior sampling yields superior enhancement quality compared to MAP-EM and deterministic inference baselines.

The remainder of this paper is structured as follows. Section 2 presents a review of related work in traditional, deep learning, and audio-visual speech enhancement methods. Section 3 discusses audio-only VAEs and introduces DVAEs. Section 4 describes the AV-RVAE model in detail. Section 5 outlines the Expectation-Maximization (EM) approach utilized in AV-RVAE. Section 6 introduces and elaborates on advanced posterior sampling algorithms. Section 7 provides comprehensive experimental results. Finally, Section 8 summarizes the paper and suggests potential future directions.

2. Related work

2.1. Traditional speech enhancement methods

Traditional speech enhancement techniques predominantly rely on statistical methods such as spectral subtraction (Boll, 1979), Wiener filtering (Lim and Oppenheim, 1979), and hidden Markov models (HMMs) (Veisi and Sameti, 2013). While effective under stationary noise and moderate signal-to-noise ratio (SNR) conditions, these methods significantly degrade in performance when encountering non-stationary noise or extremely low SNR scenarios (Vincent et al., 2018). Short-time spectral amplitude estimators (Ephraim and Malah, 1984) and log-spectral domain estimators (Ephraim and Malah, 1985) have improved upon these basic approaches by more accurately modeling speech statistics. Furthermore, non-negative matrix factorization (NMF) has become popular for modeling speech power spectral density (PSD), achieving success in controlled environments (Wilson et al., 2008; Mohammadiha et al., 2013). Despite these advancements, traditional approaches remain limited in complex, real-world acoustic conditions.

2.2. Supervised deep learning-based methods

The rise of deep learning has greatly advanced speech enhancement through supervised approaches such as deep neural networks (DNNs) and convolutional neural networks (CNNs). DNNs have significantly improved speech intelligibility and noise suppression by learning from large datasets containing paired noisy and clean speech samples (Wang et al., 2014; Xu et al., 2015). Similarly, CNN-based models effectively reduce noise while preserving speech clarity (Richter et al., 2023; Karthik and Iqbal, 2023). Despite their superior performance, supervised methods are inherently limited by their heavy reliance on labeled datasets. This reliance significantly reduces their ability to generalize to unseen noise types, creating practical challenges in real-world applications.

2.3. Unsupervised learning approaches

To address the limitations of supervised methods, unsupervised learning approaches have garnered significant attention. Variational Autoencoders (VAEs) (Kingma and Welling, 2014) have emerged prominently due to their ability to model underlying speech distributions directly from clean speech, without requiring noisy-clean paired datasets. Extensions to the standard VAE framework, including Dynamical Variational Autoencoders (DVAEs) and Recurrent Variational Autoencoders (RVAEs), leverage recurrent neural networks to model temporal dependencies more effectively (Girin et al., 2021; Leglaive et al., 2020). Although these unsupervised approaches exhibit improved generalization across noise conditions, their inference typically involves computationally intensive Expectation-Maximization (EM) procedures, limiting their practicality in resource-constrained environments.

2.4. Audio-visual speech enhancement (AVSE)

Audio-visual speech enhancement (AVSE) integrates visual features such as lip movements alongside auditory data to improve speech clarity significantly, especially under challenging noise conditions (Michelsanti et al., 2021; Gabbay et al., 2018). Visual cues help disambiguate acoustically similar phonemes, thereby enhancing intelligibility. Recent studies have demonstrated clear performance advantages of AVSE methods over audio-only models, particularly in severely noisy environments (Sadeghi et al., 2020). However, AVSE approaches encounter unique challenges, including perceptual distortions like the McGurk effect (McGurk and MacDonald, 1976) and increased computational complexity arising from multimodal integration. Recent advances have introduced adaptive fusion methods (Magrin-Chagnolleau et al., 2008; Papandreou et al., 2009) and strategies to mitigate perceptual conflicts (Getz and Toscano, 2021), underscoring the ongoing developments and potential for further enhancement.

2.5. Posterior sampling for enhanced inference

Recent advancements in posterior sampling methods offer promising solutions to improve inference accuracy in unsupervised speech enhancement frameworks. Among these, Langevin Dynamics Expectation Maximization (LDEM) and Metropolis-Adjusted Langevin Algorithm (MALA) have been specifically investigated in the context of audio-only speech enhancement (Sadeghi and Serizel, 2024). Other techniques, such as Hamiltonian Monte Carlo (HMC) (Chen and Gutmire, 2023; Neal, 2012) and Barker sampling (Livingstone and Zanella, 2022), have shown strong potential in related probabilistic modeling tasks but have not yet been applied to speech enhancement problems. These approaches leverage gradient-based updates coupled with stochastic noise injection or adaptive acceptance mechanisms to efficiently explore latent variable spaces, significantly enhancing the inference process. Additionally, LDEM and MALA have demonstrated a substantial reduction in average processing time, further increasing their viability for practical deployment. Despite these promising results, the application of advanced posterior sampling methods to multimodal AVSE frameworks remains largely unexplored. Addressing this gap presents an opportunity to significantly advance AVSE performance in real-world noisy conditions.

3. Audio-only VAE and DVAE

The Variational Autoencoder (VAE), introduced by Kingma and Welling (Kingma and Welling, 2014), is a deep generative model designed for unsupervised learning of latent variable representations. In its original form, the VAE assumes that each data point is independently and identically distributed, ignoring any temporal dependencies. While this assumption is adequate for static data, it is suboptimal for sequential data such as speech, where contextual information across time frames is critical for accurate modeling.

Let $s_t \in \mathbb{C}^F$ denote the complex-valued short-time Fourier transform (STFT) frame of clean speech at time frame t , where F is the number of frequency bins. Let $z_t \in \mathbb{R}^L$ denote the corresponding latent variable, with $L \ll F$. Over a sequence of T frames, we define $s_{1:T} = \{s_t\}_{t=1}^T$ and $z_{1:T} = \{z_t\}_{t=1}^T$.

The standard VAE defines the generative model as

$$p_\theta(s_{1:T}, z_{1:T}) = \prod_{t=1}^T p_\theta(s_t | z_t) p_\theta(z_t) \quad (1)$$

where $p_\theta(z_t) = \mathcal{N}(0, I)$ is the standard Gaussian prior and $p_\theta(s_t | z_t)$ is the likelihood parameterized by a neural decoder with parameters θ . The transformation from z_t to s_t is typically modeled by a deep neural network (DNN).

To approximate the intractable posterior $p_\theta(z_t|s_t)$, the VAE introduces a variational distribution $q_\varphi(z_t|s_t)$ as the encoder network, with parameters φ . The training objective maximizes the Evidence Lower Bound (ELBO), defined for each time frame as

$$\mathcal{L}(\theta, \varphi; s_t) = \mathbb{E}_{q_\varphi(z_t|s_t)} [\log p_\theta(s_t|z_t)] - D_{\text{KL}}(q_\varphi(z_t|s_t) \parallel p_\theta(z_t)). \quad (2)$$

This objective promotes both accurate reconstruction and latent regularization.

However, for sequential data such as speech, this frame-wise modeling is limited as it neglects dependencies across time. To address this, Dynamic Variational Autoencoders (DVAEs) (Girin et al., 2021) extend the standard VAE by incorporating temporal dependencies using recurrent neural networks (RNNs) or similar mechanisms (Golmakani et al., 2023).

In DVAEs, the generative model captures temporal dynamics via

$$p_\theta(s_{1:T}, z_{1:T}) = \prod_{t=1}^T p_\theta(s_t|s_{1:t-1}, z_{1:t}) p_\theta(z_t|s_{1:t-1}, z_{1:t-1}). \quad (3)$$

This structure allows the decoder to condition on past observations and latent states, modeling non-Markovian sequences. For $t = 1$, the boundary case is defined as

$$p_\theta(s_1|s_{1:0}, z_{1:1}) := p_\theta(s_1|z_1), \quad p_\theta(z_1|s_{1:0}, z_{1:0}) := p_\theta(z_1)$$

ensuring the model is well-defined for all time steps.

For inference, the variational posterior is

$$q_\varphi(z_{1:T}|s_{1:T}) = \prod_{t=1}^T q_\varphi(z_t|s_{1:T}, z_{1:t-1}) \quad (4)$$

with the base case $q_\varphi(z_1|s_{1:T}, z_{1:0}) := q_\varphi(z_1|s_{1:T})$. This formulation allows future observations to inform all latent variables while maintaining causal conditioning on latent history. The training of DVAEs involves maximizing ELBO over a sequence $s_{1:T}$. This can be expressed as

$$\begin{aligned} \mathcal{L}(\theta, \varphi; s_{1:T}) &= \mathbb{E}_{q_\varphi(z_{1:T}|s_{1:T})} [\log p_\theta(s_{1:T}|z_{1:T})] \\ &\quad - D_{\text{KL}}(q_\varphi(z_{1:T}|s_{1:T}) \parallel p_\theta(z_{1:T})) \end{aligned} \quad (5)$$

where

- the first term is the expected log-likelihood (reconstruction term) over the latent sequence,
- the second term is the Kullback–Leibler (KL) divergence between the approximate posterior $q_\varphi(z_{1:T}|s_{1:T})$ and the prior $p_\theta(z_{1:T})$, acting as a regularizer.

This ELBO formulation avoids directly computing the intractable marginal likelihood $\log p_\theta(s_{1:T})$, and instead optimizes a lower bound that balances reconstruction fidelity with latent space regularization.

DVAEs can be flexibly implemented using various sequence models such as BLSTMs (Bi-directional LSTM) or gated recurrent units (GRUs), enabling them to capture complex temporal dynamics and provide robust generative modeling for sequential data.

4. Audio-visual modeling with RVAE

We utilize the Audio-Visual Recurrent Variational Autoencoder (AV-RVAE) model, as described in Foroushi and Dansereau (2024), to perform audio-visual speech enhancement. During training, we use clean audio-visual pairs denoted as $\{s_t, v_t\}_{t=1}^T$, where s_t represents the audio signal at time t and v_t is a visual feature vector extracted from a lipreading network (Martinez et al., 2020). Visual information is crucial in this context as it complements the audio signal, particularly under adverse noise conditions.

The generative model follows the formulation in Leglaive et al. (2020) of

$$p_\theta(s_t|z_t, v_t) = \mathcal{N}_c(0, \text{diag}(\sigma_{\theta_s}^2(z_t, v_t))), \quad p(z_t) = \mathcal{N}(0, I) \quad (6)$$

where $\mathcal{N}_c(0, \Sigma)$ is a circularly symmetric complex Gaussian distribution. This allows the model to synthesize complex-valued STFT coefficients of speech conditioned on latent variables z_t and visual inputs v_t .

The encoder defines the approximate posterior as

$$q_\varphi(z_{1:T}|s_{1:T}, v_{1:T}) = \prod_{t=1}^T q_\varphi(z_t|z_{1:t-1}, s_{1:T}, v_{1:T}) \quad (7)$$

indicating sequential latent inference based on past latent variables and future observations.

The AV-RVAE supports both causal and non-causal variants (Girin et al., 2021; Leglaive et al., 2020). The non-causal version uses the full observation sequences $s_{1:T}, v_{1:T}$ for inference instead of $s_{1:t}, v_{1:t}$, enabling access to future context.

In line with previous work, we apply the encoder to the squared magnitude of the STFT, and learn parameters θ and φ by maximizing the ELBO.

$$\mathcal{L}(\theta, \varphi; s_{1:T}, v_{1:T}) = \sum_{t=1}^T \mathbb{E}_{q_\varphi(z_{1:t}|s_{1:T}, v_{1:T})} [\log p_\theta(s_t|z_{1:t}, v_t)]$$

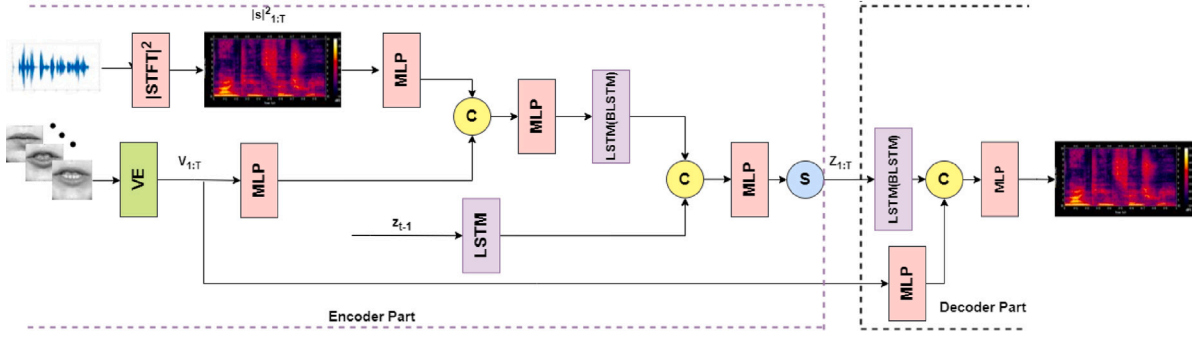


Fig. 1. Diagram illustrating the proposed AV-RVAE generative model. Key components include MLP (multilayer perceptron), LSTM (long short-term memory networks), BLSTM (Bi-directional LSTM), VE (video encoder), C (concatenation), and S (sampling in the latent space).

$$-\sum_{t=1}^T \mathbb{E}_{q_\phi(z_{1:t-1}|s_{1:T}, v_{1:T})} [D_{\text{KL}}(q_\phi(z_t|z_{1:t-1}, s_{t:T}, v_{t:T}) \parallel p(z_t))]. \quad (8)$$

This formulation balances reconstruction fidelity and prior regularization across time steps.

The model architecture is depicted in Fig. 1, which highlights its key components: MLP (multilayer perceptron), LSTM (long short-term memory), BLSTM (bidirectional LSTM), VE (video encoder), C (concatenation), and S (sampling). In the causal variant, a unidirectional LSTM is used for temporal modeling, while the non-causal version employs a bidirectional LSTM to leverage both past and future context (Girin et al., 2021).

The AV-RVAE framework leverages the complementary nature of visual information to improve speech enhancement, particularly in severely noisy environments, and provides a flexible structure for modeling sequential multimodal dependencies.

5. MAP-EM in AV-RVAE for speech enhancement

Following (Girin et al., 2021; Bie et al., 2022), we adopt a Gaussian noise model during the test phase, structured using Non-negative Matrix Factorization (NMF) (Bando et al., 2018; Leglaive et al., 2018). This model captures the structure of additive noise in audio signals. Although effective, in conditions where noise is already well approximated by a Gaussian distribution, the NMF parameterization may introduce unnecessary computational overhead.

For each time frame t , the noise component b_t is modeled as

$$p(b_t) = \mathcal{N}_c(0, \text{diag}(W h_t)), \quad (9)$$

where $W \in \mathbb{R}_+^{F \times M}$ and $H \in \mathbb{R}_+^{M \times \tilde{T}}$ contain non-negative entries, and h_t is the t th column of H . Here, T is the total number of time frames, and $\tilde{T}$ is the subset used for NMF estimation.

The observed mixture signal x_t is modeled as

$$x_t = \sqrt{g_t} s_t + b_t, \quad (10)$$

where g_t is a gain parameter scaling the clean speech. To estimate W , H , and g_t , we employ an Expectation-Maximization (EM) approach.

5.1. MAP-EM algorithm

E-step. Estimate the latent variables $z_{1:\tilde{T}}$ by maximizing the expected log-likelihood of the complete data, while keeping g_t , W , and H fixed using

$$z_{1:\tilde{T}}^* = \underset{z_{1:\tilde{T}}}{\text{argmax}} \sum_{t=1}^{\tilde{T}} \log p_{W,H}(x_t|z_{1:t}, g_t, v_t) + \log p_\theta(z_t|z_{1:t-1}, v_{t:T}). \quad (11)$$

M-step. Optimize $g_{1:\tilde{T}}$, W , and H using updated latent variables $z_{1:\tilde{T}}^*$.

• Gain update:

$$g_{1:\tilde{T}}^* = \underset{g_{1:\tilde{T}}}{\text{argmax}} \sum_{t=1}^{\tilde{T}} \log p_{W,H}(x_t|z_{1:t}^*, g_t, v_t) + \log p(g_t) \quad (12)$$

where the prior $p(g_t)$ is a Gamma distribution

$$p(g_t) = \frac{\beta^\alpha}{\Gamma(\alpha)} g_t^{\alpha-1} e^{-\beta g_t}. \quad (13)$$

This optimization can be performed using gradient ascent.

- **NMF update:**

$$W, H = \operatorname{argmax}_{W, H} \sum_{t=1}^{\tilde{T}} \log p_{W, H}(x_t | z_{1:t}^*, g_t^*, v_t) \quad (14)$$

In practice, W and H are updated using multiplicative update rules derived from the auxiliary function method, ensuring non-negativity constraints are preserved. These updates alternate between fixing W and optimizing H , and vice versa, until convergence.

5.2. Speech signal estimation

The enhanced speech signal $\hat{s}_{1:\tilde{T}}$ is estimated using the posterior mean

$$\hat{s}_{1:\tilde{T}} = \mathbb{E}_{p_{W^*, H^*}(s_{1:\tilde{T}} | x_{1:\tilde{T}}, v_{1:\tilde{T}})}[s_{1:\tilde{T}}] \approx \left\{ \frac{g_t^* \sigma_\theta^2(z_t^*, v_t)}{g_t^* \sigma_\theta^2(z_t^*, v_t) + W^* h_t^*} \odot x_t \right\}_{t=1}^{\tilde{T}} \quad (15)$$

where $\odot$ denotes element-wise multiplication. W^* and H^* are the optimized NMF parameters, and z_t^* and g_t^* are the inferred latent variables and gain parameters, respectively.

6. Posterior sampling algorithms in AV-RVAE

One of the main computational challenges of using Recurrent Variational Autoencoder (RVAE) models for unsupervised speech enhancement lies in the Expectation-Maximization (EM) algorithm. Specifically, the E-step requires iterative inference over latent variables, which becomes computationally demanding at test time. In [Foroushi and Dansereau \(2024\)](#), an EM-based approach using maximum a posteriori (MAP) estimation was employed to infer latent sequences from noisy audio-visual data.

Inspired by the posterior sampling framework proposed in [Sadeghi and Serizel \(2024\)](#), we replace the MAP optimization in the E-step with gradient-based sampling techniques. These algorithms aim to more flexibly and efficiently approximate the posterior distribution $p_\psi(z | x, v)$, where $\psi = \{W, H\}$ particularly in high-dimensional spaces. All proposed samplers utilize the *score function*

$$f_\psi(z) = \nabla_z \log p_\psi(z | x, v), \quad (16)$$

guiding proposals toward regions of high posterior density.

In the following subsections, we describe five posterior sampling algorithms: Langevin dynamics (LDEM), Metropolis-Adjusted Langevin Algorithm (MALA), Hamiltonian Monte Carlo (HMC), the Barker algorithm, and a hybrid MALA-Barker method.

6.1. Langevin dynamics expectation maximization (LDEM)

The LDEM algorithm integrates Langevin dynamics into the E-step of the EM algorithm to approximate the posterior distribution $p_\psi(z | x, v)$ through stochastic sampling. Instead of performing MAP estimation, LDEM iteratively updates latent variables by combining gradient ascent with Gaussian noise injection. This method effectively explores the posterior distribution while maintaining computational efficiency and stability.

At each iteration k , the latent variables are updated using the score function defined in Eq. (16) as

$$z^{(k)} = z^{(k-1)} + \frac{\eta}{2} f_\psi(z^{(k-1)}) + \sqrt{\eta} \zeta^{(k)}, \quad (17)$$

where $\eta > 0$ is the step size controlling the update magnitude, and $\zeta^{(k)} \sim \mathcal{N}(0, I)$ introduces Gaussian noise to promote stochastic exploration.

The update given in Eq. (17) represents a first-order discretization of Langevin dynamics. Under mild conditions, samples generated by this procedure converge asymptotically to the true posterior distribution as $k \rightarrow \infty$ ([Welling and Teh, 2011](#)). The inclusion of noise $\zeta^{(k)}$ is crucial to prevent deterministic convergence to local optima, ensuring broad posterior coverage.

The complete LDEM sampling procedure is summarized in Algorithm 1.

6.2. Metropolis-Adjusted Langevin Algorithm (MALA)

The Metropolis-Adjusted Langevin Algorithm (MALA) ([Roberts and Stramer, 2002](#); [Xifara et al., 2014](#)) is a gradient-based Markov Chain Monte Carlo (MCMC) method that extends Langevin dynamics by introducing a Metropolis–Hastings correction step. This correction step ensures convergence of the samples to the true posterior distribution, accounting for discretization errors introduced during iterative proposal updates.

At each iteration, MALA proposes a candidate latent variable via Langevin dynamics using the score function defined in Eq. (16) as

$$\tilde{z}^{(k)} = z^{(k-1)} + \frac{\eta}{2} f_\psi(z^{(k-1)}) + \sqrt{\eta} \zeta, \quad (18)$$

where $\zeta \sim \mathcal{N}(0, I)$ and $\eta > 0$ is the step size.

The proposed candidate is accepted according to the Metropolis–Hastings acceptance probability

$$\alpha = \min \left(1, \frac{p_\psi(\tilde{z}^k | x, v) q(z^{(k-1)} | \tilde{z}^k)}{p_\psi(z^{(k-1)} | x, v) q(\tilde{z}^k | z^{(k-1)})} \right), \quad (19)$$

with the proposal distribution $q(\cdot | \cdot)$ defined by the Langevin transition

$$q(u | v) \propto \exp \left(-\frac{1}{2\eta} \left\| u - v - \frac{\eta}{2} f_\psi(v) \right\|^2 \right) \quad (20)$$

The acceptance criterion in Eq. (19) corrects the bias from gradient-guided proposals, ensuring detailed balance relative to the posterior distribution.

To maintain sampling stability in high-dimensional latent spaces, the step size η is typically scaled as

$$\eta = \frac{l^2}{d_z^{1/3}}, \quad (21)$$

where d_z denotes the dimensionality of the latent variables z and l is a tunable constant, following the heuristic proposed by Roberts and Stramer (2002) to achieve optimal acceptance rates.

The complete MALA sampling procedure is summarized in Algorithm 2.

Algorithm 1 LDEM Sampler

- 1: **Input:** Initial latent variable $z^{(0)}$, step size η , number of iterations K
 - 2: **for** $k = 1$ to K **do**
 - 3: Sample noise $\zeta^{(k)} \sim \mathcal{N}(0, I)$
 - 4: Compute gradient $f_\psi(z^{(k-1)}) = \nabla_z \log p_\psi(z^{(k-1)} | x, v)$
 - 5: Update: $z^{(k)} = z^{(k-1)} + \frac{\eta}{2} f_\psi(z^{(k-1)}) + \sqrt{\eta} \zeta^{(k)}$
 - 6: **end for**
 - 7: **Output:** Sampled latent variable $z^{(K)}$
-

Algorithm 2 MALA Sampler

- 1: **Inputs:** Initial sample $z^{(0)}$, step size η , number of iterations K
 - 2: **for** each iteration $k = 1, \dots, K$ **do**
 - 3: Sample $\zeta \sim \mathcal{N}(0, I)$
 - 4: Compute proposal: $\tilde{z}^{(k)} = z^{(k-1)} + \frac{\eta}{2} f_\psi(z^{(k-1)}) + \sqrt{\eta} \zeta$
 - 5: Compute acceptance probability α using Eq. (19)
 - 6: Draw $u \sim \mathcal{U}(0, 1)$
 - 7: **if** $u < \alpha$ **then**
 - 8: Accept: $z^{(k)} = \tilde{z}^{(k)}$
 - 9: **else**
 - 10: Reject: $z^{(k)} = z^{(k-1)}$
 - 11: **end if**
 - 12: **end for**
 - 13: **Output:** Posterior samples $\{z^{(k)}\}_{k > k_{\text{burn-in}}}$
-

6.3. Hamiltonian Monte Carlo (HMC)

Hamiltonian Monte Carlo (HMC) (Neal, 2012; Chen and Ghatmirey, 2023) is a gradient-informed sampling method that introduces an auxiliary momentum variable m for each latent variable z . HMC leverages Hamiltonian dynamics from classical physics to generate proposals far from the current state while maintaining high acceptance rates. This capability significantly enhances mixing in high-dimensional spaces and mitigates the random walk behavior observed in traditional MCMC methods.

Hamiltonian system: The Hamiltonian $H(z, m)$ represents the total energy of the system with

$$H(z, m) = U(z) + K(m), \quad (22)$$

where

- $U(z) = -\log p_\psi(z | x, v)$ denotes the potential energy associated with the posterior distribution,
- $K(m) = \frac{1}{2} m^T M^{-1} m$ represents the kinetic energy, typically assuming a Gaussian prior on the momentum variable m with a mass matrix M (usually set to I).

Hamiltonian dynamics: These dynamics follow the differential equations

$$\begin{aligned}\frac{dz}{dt} &= \nabla_m H(z, m) = M^{-1} m, \\ \frac{dm}{dt} &= -\nabla_z H(z, m) = -\nabla_z U(z),\end{aligned}\tag{23}$$

which conserve total energy $H(z, m)$, guiding trajectories through high-probability regions of the posterior distribution.

Leapfrog integration: To numerically approximate the continuous Hamiltonian dynamics, HMC employs the leapfrog integrator. Beginning from $(z^{(k-1)}, m^{(k-1)})$, the leapfrog method iteratively updates positions and momenta over L steps with a fixed step size ϵ :

$$m^{(l+1/2)} = m^{(l)} - \frac{\epsilon}{2} \nabla_z U(z^{(l)}),\tag{24}$$

$$z^{(l+1)} = z^{(l)} + \epsilon M^{-1} m^{(l+1/2)},\tag{25}$$

$$m^{(l+1)} = m^{(l+1/2)} - \frac{\epsilon}{2} \nabla_z U(z^{(l+1)}).\tag{26}$$

Metropolis correction: Upon completing the leapfrog integration, the resulting proposal $(\tilde{z}, \tilde{m})$ is accepted with probability

$$\alpha = \min \left(1, \exp \left(H(z^{(k-1)}, m^{(k-1)}) - H(\tilde{z}, \tilde{m}) \right) \right),\tag{27}$$

ensuring detailed balance and correct posterior convergence. The complete procedure is outlined in Algorithm 3.

Algorithm 3 HMC Sampler

```

1: Input: Initial position  $z^{(0)}$ , step size  $\epsilon$ , leapfrog steps  $L$ , total iterations  $K$ 
2: for  $k = 1$  to  $K$  do
3:   Sample momentum  $m^{(k-1)} \sim \mathcal{N}(0, M)$ 
4:   Set  $(\tilde{z}, \tilde{m}) \leftarrow (z^{(k-1)}, m^{(k-1)})$ 
5:   for  $l = 1$  to  $L$  do
6:     {Leapfrog steps}  $\tilde{m} \leftarrow \tilde{m} - \frac{\epsilon}{2} \nabla_z U(\tilde{z})$ ,  $\tilde{z} \leftarrow \tilde{z} + \epsilon M^{-1} \tilde{m}$ ,  $\tilde{m} \leftarrow \tilde{m} - \frac{\epsilon}{2} \nabla_z U(\tilde{z})$ 
7:   end for
8:   Compute acceptance probability using Eq. (27)
9:   Sample  $u \sim \mathcal{U}(0, 1)$ 
10:  if  $u < \alpha$  then
11:     $z^{(k)} \leftarrow \tilde{z}$ 
12:  else
13:     $z^{(k)} \leftarrow z^{(k-1)}$ 
14:  end if
15: end for
16: Output:  $z^{(K)}$ 
```

Remarks: The number of leapfrog steps L and the step size ϵ are hyperparameters that balance computational cost and acceptance rate. Empirically, ϵL is often set to a constant (e.g., 1), and acceptance rates between 60%–80% are typically targeted.

6.4. Barker algorithm

The Barker algorithm (Livingstone and Zanella, 2022) is a variant of Metropolis–Hastings that employs a logistic-based acceptance rule. Unlike traditional schemes that use a hard acceptance threshold, Barker’s smoother acceptance probability enhances mixing, particularly in complex or multimodal posteriors.

Let $z^{(k-1)}$ be the current sample, and propose $\tilde{z}^{(k)}$ using a Langevin-style update

$$\tilde{z}^{(k)} = z^{(k-1)} + \eta f_\psi(z^{(k-1)}) + \sqrt{\eta} \zeta, \quad \zeta \sim \mathcal{N}(0, I),\tag{28}$$

where η is the step size and $f_\psi(z)$ is defined in Eq. (16).

Acceptance criterion: The proposed sample is accepted with probability

$$\alpha = \frac{p_\psi(\tilde{z}^{(k)} \mid x, v)}{p_\psi(\tilde{z}^{(k)} \mid x, v) + p_\psi(z^{(k-1)} \mid x, v)},\tag{29}$$

providing a smoother transition rule compared to Metropolis–Hastings. For example:

- If $p_\psi(\tilde{z}^{(k)} \mid x, v) \gg p_\psi(z^{(k-1)} \mid x, v)$, then $\alpha \rightarrow 1$,
- If $p_\psi(\tilde{z}^{(k)} \mid x, v) \ll p_\psi(z^{(k-1)} \mid x, v)$, then $\alpha \rightarrow 0$,
- If they are similar, $\alpha \approx 0.5$, encouraging probabilistic transitions.

The complete procedure is outlined in Algorithm 4.

Remarks:

- The Barker acceptance function reduces abrupt transitions and rejection rates in complex posteriors.
- It is less aggressive than Metropolis–Hastings in favoring high-probability regions, which may slow convergence.
- As with all Langevin-type samplers, careful tuning of η is essential for balancing exploration and stability.

Algorithm 4 Barker Sampler

```

1: Input: Initial sample  $z^{(0)}$ , step size  $\eta$ , total iterations  $K$ 
2: for  $k = 1$  to  $K$  do
3:   Sample noise  $\zeta \sim \mathcal{N}(0, I)$ 
4:   Compute proposal:  $\tilde{z}^{(k)} = z^{(k-1)} + \eta f_\psi(z^{(k-1)}) + \sqrt{\eta} \zeta$ 
5:   Compute acceptance probability using Eq. (29)
6:   Sample  $u \sim \mathcal{U}(0, 1)$ 
7:   if  $u < \alpha$  then
8:     Accept:  $z^{(k)} = \tilde{z}^{(k)}$ 
9:   else
10:    Reject:  $z^{(k)} = z^{(k-1)}$ 
11:   end if
12: end for
13: Output:  $\{z^{(k)}\}_{k > k_{\text{burn-in}}}$ 

```

6.5. Hybrid MALA-Barker algorithm

The Hybrid MALA-Barker algorithm combines the strengths of the Metropolis-Adjusted Langevin Algorithm (MALA) and the Barker acceptance rule. It aims to exploit the rapid convergence properties of MALA during the early sampling phase and the smooth, adaptive behavior of Barker sampling in the later phase.

This two-stage approach enables the sampler to quickly reach high-density regions of the posterior and then robustly explore them without suffering from sharp acceptance thresholds.

Let $z^{(k-1)}$ be the latent variable at iteration $(k - 1)$, and let $f_\psi(z)$ be the score function (see Eq. (16)). The hybrid algorithm operates as follows:

- For iterations $k \leq K_{\text{MALA}}$: use MALA proposals and Metropolis–Hastings acceptance.
- For iterations $k > K_{\text{MALA}}$: switch to Barker proposals and logistic acceptance.

Step 1: MALA phase. In the MALA phase, proposals are generated by

$$\tilde{z}^{(k)} = z^{(k-1)} + \frac{\eta}{2} f_\psi(z^{(k-1)}) + \sqrt{\eta} \zeta, \quad \zeta \sim \mathcal{N}(0, I). \quad (30)$$

The acceptance probability is

$$\alpha_{\text{MALA}} = \min \left(1, \frac{p_\psi(\tilde{z}^{(k)} | x, v) q(z^{(k-1)} | \tilde{z}^{(k)})}{p_\psi(z^{(k-1)} | x, v) q(\tilde{z}^{(k)} | z^{(k-1)})} \right), \quad (31)$$

where $q(\cdot | \cdot)$ denotes the Langevin proposal distribution (20).

Step 2: Barker phase. For $k > K_{\text{MALA}}$, the proposal is

$$\tilde{z}^{(k)} = z^{(k-1)} + \eta f_\psi(z^{(k-1)}) + \sqrt{\eta} \zeta, \quad (32)$$

with acceptance probability

$$\alpha_{\text{Barker}} = \frac{p_\psi(\tilde{z}^{(k)} | x, v)}{p_\psi(\tilde{z}^{(k)} | x, v) + p_\psi(z^{(k-1)} | x, v)}. \quad (33)$$

Algorithm 5 provides a complete summary of the HMC sampling process.

Remarks:

- The MALA phase promotes rapid convergence through gradient-guided proposals with MH correction.
- The Barker phase enhances exploration by permitting smoother transitions between states.
- The switch threshold K_{MALA} can be tuned based on observed mixing behavior.
- This hybrid design is especially useful in high-dimensional latent spaces where a single sampler may be insufficient.

Table 1

Speech-enhancement results for audio-only (A-RVAE) and audio-visual (AV-RVAE) models using different posterior sampling methods. Metrics are reported across SNRs $\{-5, 0, 5, 10, 15\}$, and their averages are shown in the rightmost column.

Metrics	SI-SDR (dB)					PESQ						STOI						
	SNR (dB)					Avg.	SNR (dB)					Avg.	SNR (dB)					Avg.
	-5	0	5	10	15		-5	0	5	10	15		-5	0	5	10	15	
Input	-12.33	-7.56	-2.88	2.11	7.35	-2.66	1.51	1.77	2.07	2.39	2.88	2.12	0.19	0.30	0.43	0.57	0.68	0.43
Audio-only model (A-RVAE)																		
MAp-EM	-4.81	0.22	5.92	10.83	15.25	5.48	1.95	2.23	2.61	2.85	3.07	2.54	0.24	0.38	0.53	0.68	0.76	0.51
LDEM	-4.10	1.22	6.48	10.83	15.31	5.95	1.81	2.10	2.44	2.70	2.93	2.40	0.22	0.34	0.46	0.62	0.69	0.47
MALA	-3.50	1.65	7.11	11.55	15.89	6.54	1.91	2.19	2.53	2.79	3.00	2.48	0.26	0.40	0.54	0.69	0.75	0.53
MALA (opt. η)	-3.42	1.62	7.08	11.50	15.81	6.52	1.82	2.08	2.40	2.68	2.90	2.38	0.27	0.40	0.53	0.67	0.74	0.52
HMC	-3.73	1.46	6.89	11.10	15.60	6.67	1.83	2.05	2.37	2.64	2.88	2.35	0.25	0.38	0.51	0.65	0.72	0.50
BARKER	-3.68	1.55	6.85	11.32	15.72	6.1	1.80	2.05	2.38	2.66	2.88	2.35	0.21	0.35	0.46	0.63	0.71	0.47
BARKER+MALA	-3.20	1.89	7.41	11.73	16.05	6.78	1.89	2.13	2.47	2.74	2.96	2.44	0.26	0.39	0.52	0.66	0.73	0.51
Audio-visual model (AV-RVAE)																		
MAP-EM	-3.25	1.91	7.32	11.68	16.16	6.76	2.02	2.31	2.66	2.91	3.09	2.60	0.27	0.41	0.54	0.69	0.77	0.54
LDEM	-3.02	2.13	7.57	11.95	16.41	6.81	2.01	2.29	2.65	2.91	3.08	2.59	0.26	0.41	0.54	0.68	0.75	0.53
MALA	-2.81	2.31	7.81	12.16	16.63	7.22	2.03	2.36	2.72	2.98	3.15	2.65	0.27	0.42	0.55	0.71	0.78	0.55
MALA (opt. η)	-2.83	2.29	7.78	12.13	16.61	7.20	1.94	2.23	2.58	2.82	3.01	2.52	0.29	0.43	0.56	0.71	0.78	0.55
HMC	-3.08	2.08	7.53	11.91	16.35	6.96	1.96	2.24	2.59	2.84	3.02	2.53	0.29	0.43	0.56	0.72	0.79	0.56
BARKER	-3.67	1.74	6.91	11.47	15.87	6.46	2.03	2.33	2.67	2.93	3.12	2.62	0.22	0.36	0.48	0.65	0.72	0.49
BARKER+MALA	-2.59	2.53	8.11	12.41	16.93	7.48	2.11	2.42	2.77	3.02	3.21	2.71	0.29	0.44	0.57	0.72	0.79	0.56

Table 2

RTF (average processing time per 1-s speech) values.

Method	MAP-EM	LDEM	MALA	MALA with opt. η	HMC	Barker	BARKER+MALA
Value	119.93	24.10	33.65	34.05	117.73	32.92	33.65

Algorithm 5 Hybrid MALA-Barker Sampler

```

1: Input: Initial sample  $z^{(0)}$ , step size  $\eta$ , MALA threshold  $K_{\text{MALA}}$ , total iterations  $K$ 
2: for  $k = 1$  to  $K$  do
3:   Sample noise  $\zeta \sim \mathcal{N}(0, I)$ 
4:   if  $k \leq K_{\text{MALA}}$  then
     {MALA phase} Propose  $\tilde{z}^{(k)}$  using Eq. (30) Compute  $\alpha_{\text{MALA}}$  using Eq. (31) Set  $\alpha \leftarrow \alpha_{\text{MALA}}$ 
5:   else
     {Barker phase} Propose  $\tilde{z}^{(k)}$  using Eq. (32) Compute  $\alpha_{\text{Barker}}$  using Eq. (33) Set  $\alpha \leftarrow \alpha_{\text{Barker}}$ 
6:   end if
7:   Sample  $u \sim \mathcal{U}(0, 1)$ 
8:   if  $u < \alpha$  then
9:      $z^{(k)} \leftarrow \tilde{z}^{(k)}$ 
10:  else
11:     $z^{(k)} \leftarrow z^{(k-1)}$ 
12:  end if
13: end for
14: Output:  $\{z^{(k)}\}_{k > k_{\text{burn-in}}}$ 

```

7. Experimental results

This section presents and discusses the results of audio-visual speech enhancement using AV-RVAE (Foroushi and Dansereau, 2024) with several posterior sampling algorithms, namely LDEM, MALA, HMC, and Barker, as well as the EM-based algorithm from Foroushi and Dansereau (2024). In addition, as an ablation study, we report results for an audio-only baseline (A-RVAE), obtained by disabling the visual branch of the model while keeping the same data, training protocol, and inference budgets as AV-RVAE. This ablation study isolates the specific contribution of visual fusion to enhancement performance. All experiments are conducted on the NTCD-TIMIT dataset within an unsupervised enhancement framework that combines a recurrent variational autoencoder (RVAE) with a non-negative matrix factorization (NMF) noise model.

7.1. Evaluation metrics

We evaluate enhancement quality using three objective metrics:

- **SI-SDR** (Roux et al., 2019): Measures signal distortion; higher is better.
- **PESQ** (Rix et al., 2001): Evaluates perceived quality from -0.5 to 4.5 .
- **STOI** (Taal et al., 2011): Assesses intelligibility on a $0-1$ scale.

We also report the Real-Time Factor (RTF), defined as the total runtime divided by the signal duration (unitless). Lower RTF implies higher computational efficiency. All experiments were performed on an NVIDIA Tesla A100 GPU.

7.2. AV-RVAE architecture

Our model uses a visual encoder (VE) based on a 3D CNN-ResNet from Martinez et al. (2020), fixed during training. The encoder uses bidirectional LSTM (BLSTM) layers (128 hidden units), followed by a multilayer perceptron (MLP). The decoder reconstructs frames via an MLP using latent sequences processed through another BLSTM as shown in Fig. 1.

The model uses a 16-dimensional latent space and 513-point spectral input. Training employs the Adam optimizer (Kingma and Ba, 2014) with a learning rate of 0.001, batch size 128, and early stopping with 50-epoch patience. A-RVAE is defined as the same RVAE architecture and training procedure without visual conditioning (i.e., the VE is disabled and the latent prior/encoder are not conditioned on visual features).

The AV-RVAE employs bidirectional LSTMs (BLSTMs) in both encoder and decoder, enabling the use of full temporal context. Consequently, the current implementation is non-causal and operates in an offline setting. This design maximizes enhancement quality but is incompatible with real-time or streaming applications. Developing a causal variant, for instance using unidirectional LSTMs with the proposed MALA-based inference, is left for future work.

7.3. Experimental setup

We evaluate all sampling methods over 100 EM iterations. The sampling step size is set to 0.005 (except for MALA with optimized η , computed via Eq. (21) with $l = 0.005$). We use $K = 1$ for EM and LDEM, and $K = 10$ (burn-in = 5) for MALA, HMC, and Barker. The hybrid MALA+Barker uses $K_{\text{MALA}} = 10$.

For the Hamiltonian Monte Carlo (HMC) sampler, the leapfrog integrator used $L = 10$ steps per iteration with a step size of $\epsilon = 0.05$, and a diagonal mass matrix $M = \mathbf{I}$. The average acceptance rates observed across the evaluation set were 0.63 for MALA, 0.78 for HMC, and 0.67 for Barker, indicating stable and well-tuned sampling behavior. These acceptance rates were measured as the proportion of accepted proposals after burn-in and averaged over all test utterances.

NMF parameters are initialized randomly (non-negative) and kept consistent across runs (Girin et al., 2021), with $M = 8$. Gamma parameters α and β are set to 1. The RVAE sequence length is $T = 50$. A-RVAE uses the same data splits, preprocessing, training schedule, and inference budgets as AV-RVAE, ensuring a controlled comparison of visual fusion. We built our implementation upon the publicly available posterior sampling code from Sadeghi and Serizel (2024),¹ which served as a foundation for extending the methods to our audio-visual RVAE framework. Our implementation may be made available upon request.

To enable fair runtime comparison across methods, we log “anytime” curves that record SI-SDR after each EM iteration along with cumulative wall-clock time. SI-SDR is selected for this analysis because it directly measures signal reconstruction accuracy and is the most sensitive to convergence behavior among the considered metrics.

During inference, we log anytime curves by recording SI-SDR after every EM iteration together with the cumulative wall-clock time. This enables fair comparisons of (i) how quickly each sampler improves quality under a fixed time budget and (ii) the final quality at convergence. Unless stated otherwise, all samplers are run with the same maximum of 100 EM iterations on the same hardware (NVIDIA Tesla A100) and software stack (PyTorch, float32). For practicality, we employ early stopping on validation SI-SDR with patience $p=3$ iterations and minimum improvement $\delta=0.05$ dB; when triggered, we report the stopping iteration and the corresponding wall-clock time.

7.4. Dataset

For training, validation, and testing, we used the NTCD-TIMIT dataset (Abdelaziz, 2017), which contains audiovisual recordings from 56 English speakers. These speakers were split into 39 for training, 8 for validation, and 9 for testing, with each participant uttering a unique set of 98 sentences. In total, the dataset includes 5488 audiovisual recordings, each clip lasting approximately 5 s.

The visual data are composed of videos focused on the lip region of interest (ROI), captured at a rate of 30 frames per second, with each frame sized at 67×67 pixels. To synchronize visual and audio modalities, the lip ROI frames were temporally upsampled, aligning the visual frame rate to that of audio processing. The audio data was sampled at 16 kHz, with spectral features extracted via short-time Fourier transform (STFT) using a 64 ms window (1024 samples per frame) and 75% overlap, yielding 513 frequency bins.

¹ Available at <https://github.com/msaadeghi/ps-rvae-se>.

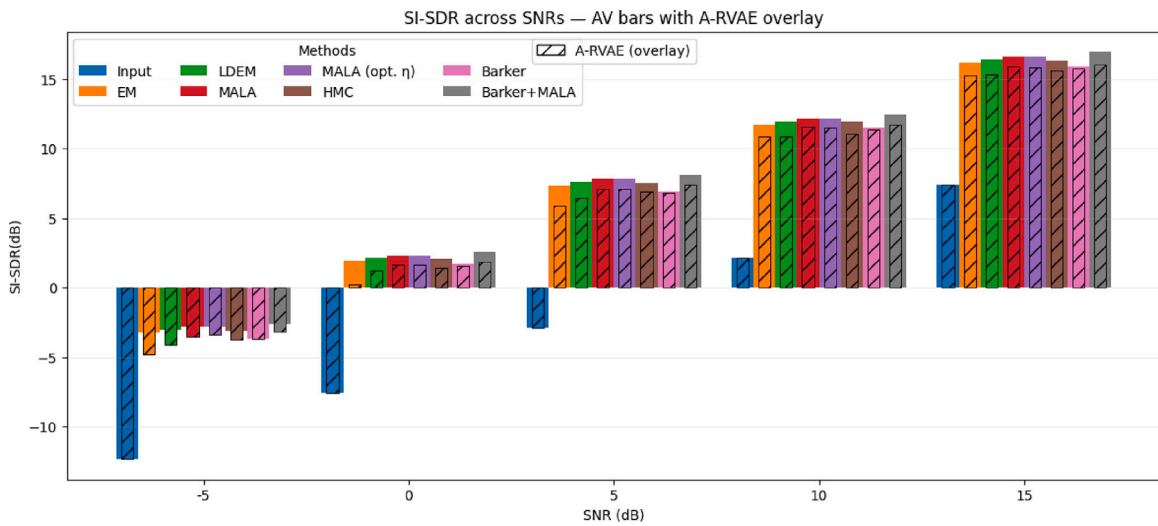


Fig. 2. SI-SDR scores across SNR levels for AV-RVAE (solid) and A-RVAE (hatched) across different sampling methods.

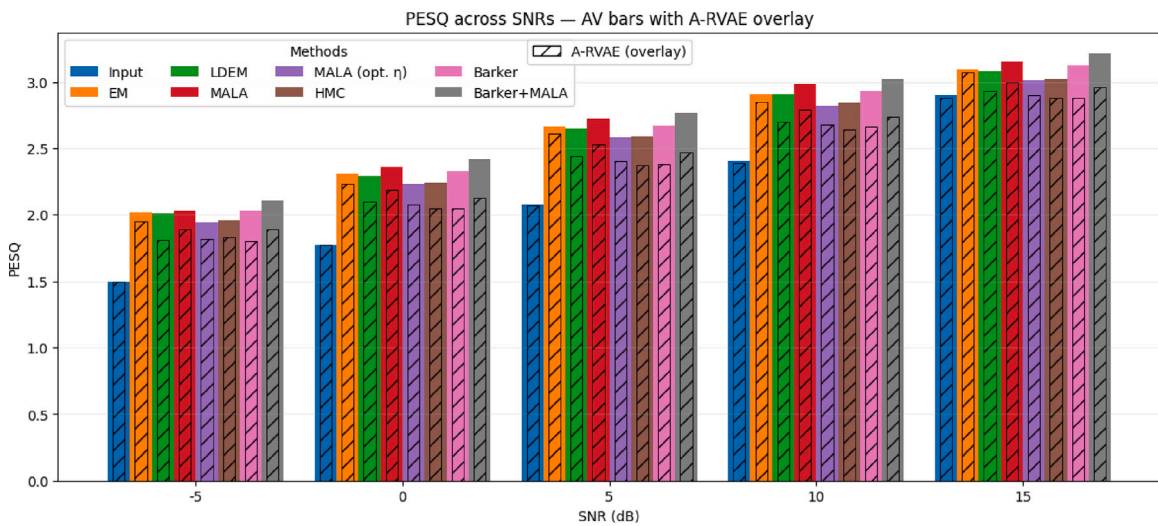


Fig. 3. PESQ scores across SNR levels for AV-RVAE (solid) and A-RVAE (hatched) across different sampling methods.

To evaluate speech enhancement (SE) performance, we used the noisy speech samples from the NTCD-TIMIT dataset (Abdelaziz, 2017). This noisy dataset incorporates six types of noise: Living Room, White, Cafe, Car, Babble, and Street; each presented at five distinct signal-to-noise ratios (SNRs): -5 , 0 , 5 , 10 , and 15 dB. To create the test set, we randomly selected four utterances for each noise type and SNR level per speaker, resulting in a total of 1080 noisy test samples across the dataset. This comprehensive range of noise conditions provides a robust basis for evaluating the performance of the SE model in diverse real-world acoustic environments.

7.5. Results and analysis

Table 1 summarizes speech-enhancement performance across different SNRs for both the audio-only (A-RVAE) and the proposed audio-visual (AV-RVAE) models, evaluated under seven posterior inference strategies: MAP-Expectation Maximization (MAP-EM), Langevin Dynamics EM (LDEM), Metropolis-Adjusted Langevin Algorithm (MALA), MALA with optimized step size (η), Hamiltonian Monte Carlo (HMC), Barker, and the hybrid Barker+MALA. Figs. 2–4 present SI-SDR, PESQ, and STOI results averaged over all noise types. In each plot, solid bars denote AV-RVAE performance, while the hatched overlays represent the corresponding A-RVAE results, facilitating direct visual comparison. Here, the A-RVAE serves as an ablation study that isolates the effect of removing visual conditioning, thereby quantifying the specific contribution of the visual modality to overall enhancement performance. The rightmost columns of Table 1 report the average performance of each inference method, providing a more aggregated comparison across conditions.

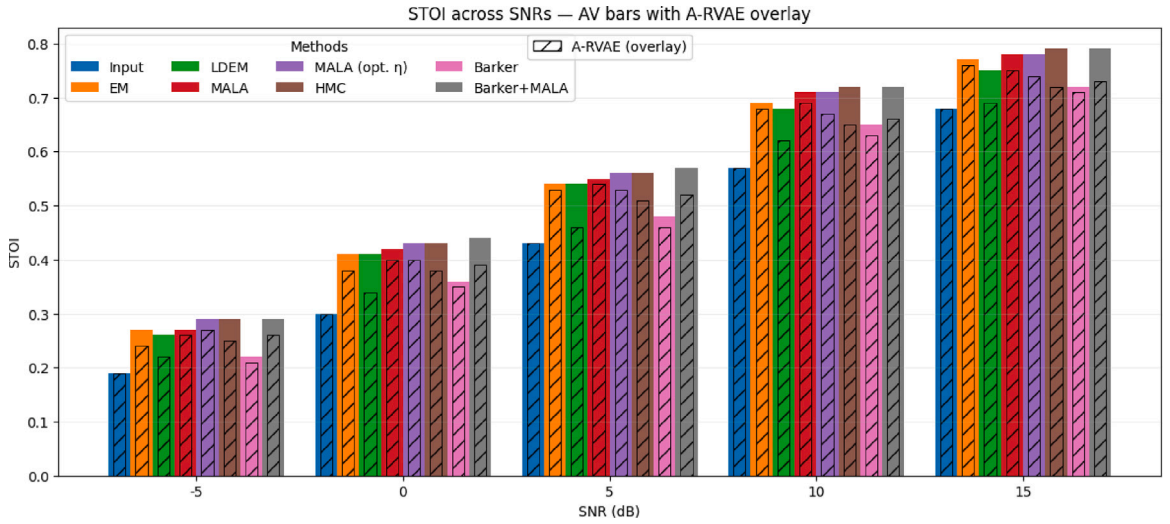


Fig. 4. STOI scores across SNR levels for AV-RVAE (solid) and A-RVAE (hatched) across different sampling methods.

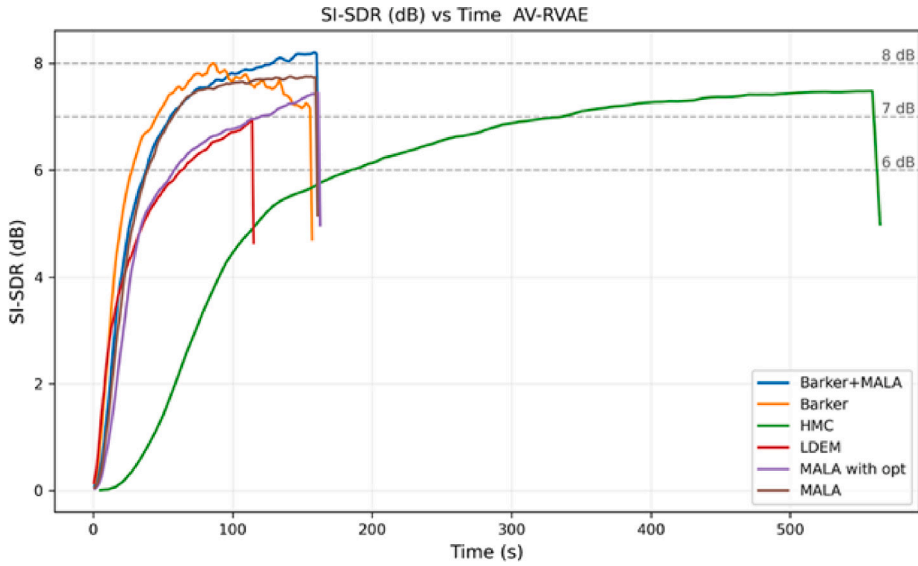


Fig. 5. Anytime enhancement performance of all samplers with AV-RVAE, showing SI-SDR (dB) versus wall-clock time. Lighter-weight samplers (Barker, MALA) reach 7 dB more rapidly, while HMC improves more gradually. Each curve terminates when the fixed iteration budget is reached.

Across all inference schemes and SNR levels, the AV-RVAE consistently outperforms the audio-only baseline in perceptual quality and intelligibility, as reflected in the average SI-SDR, PESQ, and STOI scores. These gains are particularly pronounced at low SNRs (−5 dB and 0 dB), where the visual stream provides valuable complementary information to the degraded audio signal. The results confirm that visual cues substantially improve robustness under challenging noise conditions.

Among the tested inference algorithms, the hybrid Barker+MALA achieves the highest overall performance, yielding the best average SI-SDR (7.48 dB) and PESQ (2.71), while sharing the top STOI (0.56) with HMC. Plain MALA performs comparably in distortion reduction in SI-SDR and maintains strong perceptual metrics, indicating that the Barker correction further enhances asymptotic reconstruction quality. The HMC method performs comparably in intelligibility but incurs significantly higher computational cost. Meanwhile, LDEM and MAP-EM exhibit stable convergence but lower overall enhancement quality, reflecting the benefits of gradient-informed sampling in MALA-based methods.

Fig. 5 plots SI-SDR versus wall-clock time to illustrate the “anytime” behavior of each sampler, i.e., how enhancement quality evolves as inference progresses. Lightweight samplers such as Barker, MALA, MALA+Barker and MALA (opt. η) rapidly reach usable quality levels (around 6–7 dB) with low latency, while HMC continues to improve gradually but at a higher runtime cost. In contrast, LDEM exhibits slower early gains, reflecting its more limited exploration efficiency. To further quantify convergence behavior, Fig.

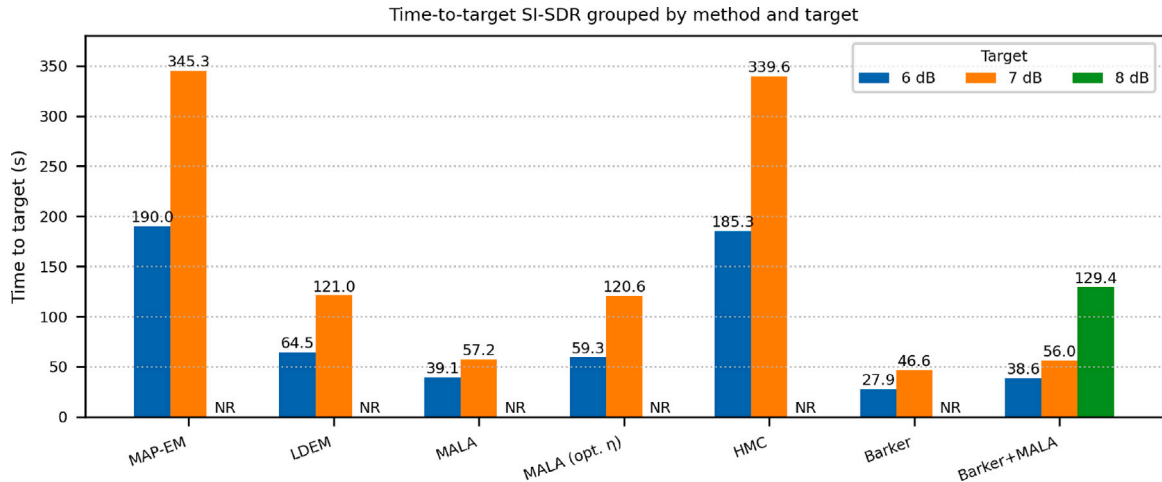


Fig. 6. Time-to-target analysis for different posterior sampling methods. Bars show the wall-clock time required to reach SI-SDR targets of 6, 7, and 8 dB, using a common vertical axis for direct comparison. Barker and Barker+MALA converge fastest to 6–7 dB, and only Barker+MALA attains 8 dB within the fixed iteration budget (others are marked NR).

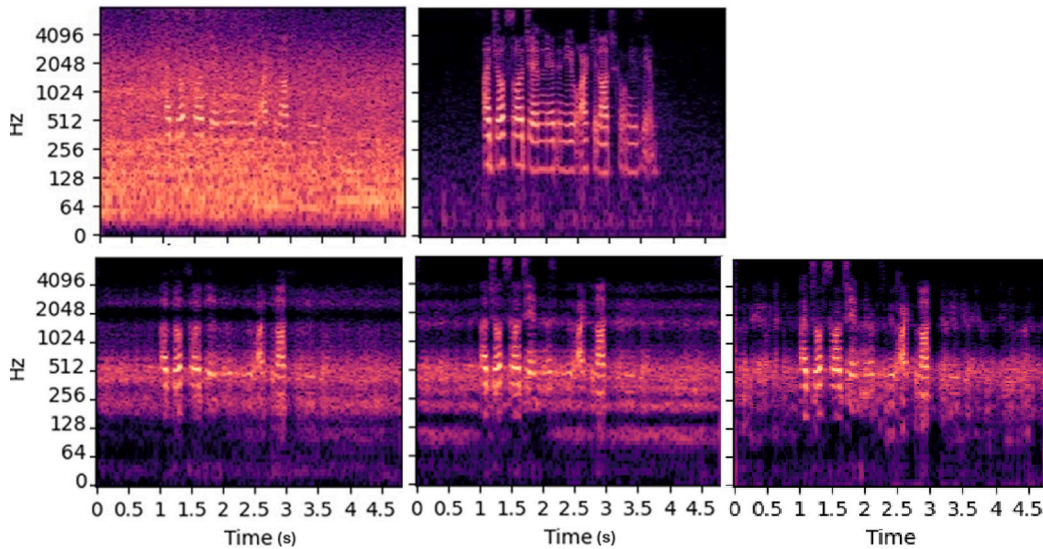


Fig. 7. Example spectrogram results: noisy speech (upper-left), clean speech (upper-right), A-RVAE (bottom-left), AV-RVAE with MAP-EM (bottom-middle) and AV-RVAE with MALA+BARKER posterior-sampling result (bottom-right).

6 reports the *time-to-target* for SI-SDR thresholds of 6, 7, and 8 dB. Barker and Barker+MALA achieve 6–7 dB fastest, and only the hybrid Barker+MALA reaches 8 dB within the fixed time budget. All curves correspond to input mixtures at 5 dB SNR, representative of moderate noise conditions .

Table 2 summarizes the computational cost in terms of the Real-Time Factor (RTF; lower is better). LDEM exhibits the lowest RTF (24.10), followed by a cluster of efficient stochastic samplers. Barker, MALA, MALA (opt. η), and Barker+MALA—all around 33–34. MAP-EM and HMC remain significantly slower (approximately 118–120). When combined with the anytime curves, these results demonstrate a clear efficiency–quality trade-off: LDEM achieves the fastest runtime but saturates at lower quality, whereas Barker+MALA and MALA deliver superior enhancement at moderate computational cost.

Overall, audio-visual fusion consistently improves all metrics across sampling strategies, confirming the benefit of integrating visual cues for robust enhancement. Among inference methods, Barker+MALA achieves the best balance between quality, stability, and computational efficiency, whereas LDEM remains suitable for latency-critical applications.

Fig. 7 provides a qualitative comparison of spectrograms for a representative utterance at 5 dB SNR. The noisy input (top left) shows strong broadband interference that masks speech harmonics. The MAP-EM-based A-RVAE (bottom left) removes some noise but leaves residual smearing, while the AV-RVAE (bottom center) achieves cleaner harmonic contours owing to the inclusion of visual cues. The proposed AV-RVAE with MALA+Barker (bottom right) produces the most structured and intelligible spectrogram, closely matching the clean reference (top right) with minimal distortion and enhanced harmonic continuity. These visual observations align with the quantitative gains in Table 1, confirming that advanced posterior sampling combined with visual fusion leads to more accurate and perceptually natural speech reconstructions.

8. Conclusion

This study investigated advanced posterior sampling for audio-visual speech enhancement using a Recurrent Variational Autoencoder (AV-RVAE). By jointly exploiting audio and visual cues, the proposed approach improves speech quality, intelligibility, and robustness under challenging noise conditions. An ablation study using the audio-only baseline (A-RVAE) confirmed the critical role of the visual modality, as AV-RVAE consistently achieved higher SI-SDR, PESQ, and STOI scores, particularly at low SNRs, demonstrating the effectiveness of visual fusion for noise-robust enhancement.

Among the posterior sampling strategies evaluated, the hybrid MALA+Barker algorithm provided the best overall performance, achieving the highest average SI-SDR, PESQ, and STOI across SNR levels. The plain MALA sampler also performed strongly in distortion reduction, while LDEM and step-size-optimized MALA achieved favorable trade-offs between computational efficiency and perceptual quality.

Integrating sampling-based posterior inference within the AV-RVAE framework substantially enhanced latent variable estimation and overall enhancement effectiveness. As the current architecture employs bidirectional LSTMs, it operates in an offline (non-causal) setting. Future work will explore causal variants and lightweight noise modeling for real-time AV-RVAE inference.

CRedit authorship contribution statement

Z. Foroushi: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation. **R.M. Dansereau:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT 4 to enhance the clarity and quality of the English language presentation. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Zohre Foroushi reports financial support, article publishing charges, and travel were provided by Natural Sciences and Engineering Research Council of Canada. Richard Dansereau reports financial support, article publishing charges, and travel were provided by Natural Sciences and Engineering Research Council of Canada. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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